

Установка Autodock-Vina

```
sudo apt install autodock-vina
```

проверка установки:

```
vina --version
```

(Данная версия будет недоступна для работы в конде!!!)

Установка conda (+)

```
cd /tmp
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86\_64.sh
chmod +x Miniconda3-latest-Linux-x86_64.sh
./Miniconda3-latest-Linux-x86_64.sh
source ~/.bashrc
conda --version
```

Установка autodock-vina в окружении конды (+)

```
conda create -n vina python=3
conda activate vina
conda config --env --add channels conda-forge
conda install -c conda-forge numpy swig boost-cpp libboost sphinx sphinx_rtd_theme
pip install vina
```

Установка Meeko внутри конды (+)

```
conda activate vina
conda install python=3.10
conda install -c conda-forge numpy scipy rdkit vina meeko gemmi autogrid
pip install prody
```

Установка Meeko

```
pip install -U numpy scipy rdkit vina meeko gemmi prody
```

тут у меня возникла ошибка:

```
esh@debian: ~  
sion `CXXABI_1.3.15' not found (required by /tmp/tmpxkz_yu8d/build/py/gemmi/gemmi_ext.cpython-314-x86_64-linux-gnu.so)  
ninja: build stopped: subcommand failed.  
  
*** CMake build failed  
[end of output]  
  
note: This error originates from a subprocess, and is likely not a problem with pip.  
ERROR: Failed building wheel for gemmi  
Building wheel for prody (pyproject.toml) ... done  
Created wheel for prody: filename=prody-2.6.1-cp314-cp314-linux_x86_64.whl size=36291242 sha256=4bc7b3fef6a8027c0471bee3a81299ed6dc1e6fd121b94397abd38e308d48aea  
Stored in directory: /home/esh/.cache/pip/wheels/6f/1e/17/1dc1199ae8749ce1ba9376db009a2e5146fcad06a149822df1  
Successfully built prody  
Failed to build gemmi  
error: failed-wheel-build-for-install  
  
x Failed to build installable wheels for some pyproject.toml based projects  
↳ gemmi  
(vina) esh@debian:~$
```

Если возникнет такая же ошибка, нужно выполнить команду `conda install -c conda-forge sxx-compiler libstdcxx-ng`, она устанавливает нужные библиотеки для компилятора, затем снова команду `pip install -U numpy scipy rdkit vina meeko gemmi prody`

Проверяем установку:

`mk_prepare_ligand.py`

Если все успешно работает, будет следующий вывод:

```
(vina) esh@debian:~$ mk_prepare_ligand.py  
usage: mk_prepare_ligand.py [-h] [-v] [-i INPUT_MOLECULE_FILENAME] [--name_from_prop NAME_FROM_PROP] [--o OUTPUT_PDBQT_FILENAME] [--multimol_outdir MULTIMOL_OUTPUT_DIR] [--multimol_prefix MULTIMOL_PREFIX] [-z] [--multimol_targz_size MULTIMOL_TARGZ_SIZE] [-] [-c CONFIG_FILE] [--rigid_macrocycles] [--macrocycle_allow_A] [--keep_chorded_rings] [--keep_equivalent_rings] [--min_ring_size MIN_RING_SIZE] [-w] [--merge_these_atom_types [MERGE_THESE_ATOM_TYPES ...]] [-r SMARTS] [-b i j [i j ...]] [-a] [-p LOAD_ATOM_PARAMS [LOAD_ATOM_PARAMS ...]] [--aa [{"smarts": "<smarts pattern>", "atype": "atomtype name"}, {"smarts": "<smarts pattern>", "atype": "atomtype name"}]] [--double_bond_penalty DOUBLE_BOND_PENALTY] [--charge_model {gasteiger,espaloma,zero,read}] [--charge_atom_prop CHARGE_ATOM_PROP] [--bad_charge_ok] [--add_index_map] [--remove_smiles] [--rename_atoms] [--reactive_smarts REACTIVE_SMARTS] [--reactive_smarts_idx REACTIVE_SMARTS_IDX] [--receptor RECEPTOR] [--rec_residue REC_RESIDUE] [--tether_smarts TETHER_SMARTS] [--tether_smarts_indices IDX IDX]  
mk_prepare_ligand.py: error: the following arguments are required: -i/--mol  
(vina) esh@debian:~$
```

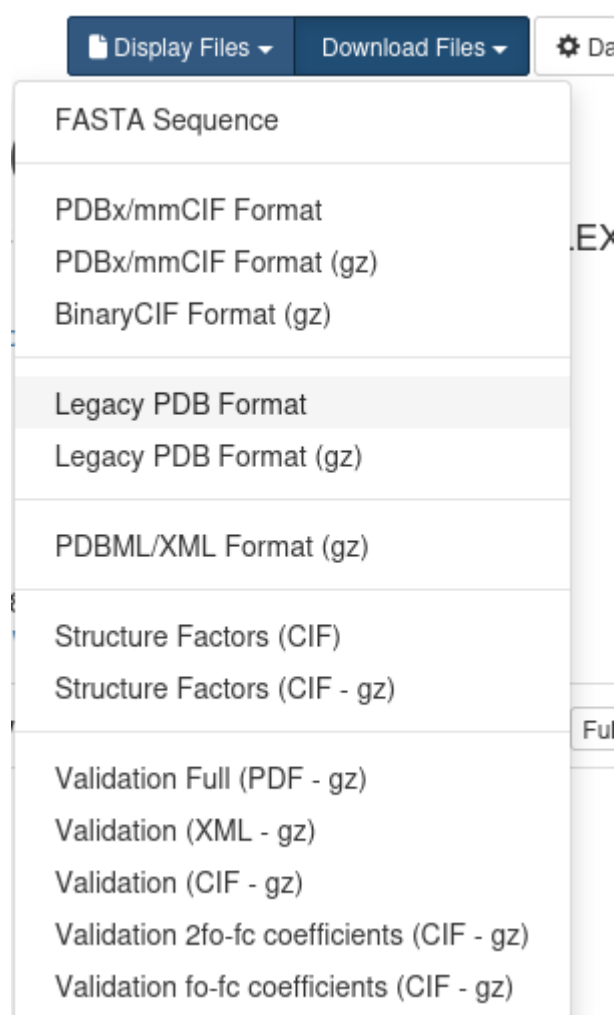
После закрытия терминала для начала работы с vina нужно сделать следующее:

1. активировать среду `conda activate vina`
2. для проверки запуска вины ввести `vina`. Если всё хорошо, будет вывод:

```
esh@debian: ~  
[--receptor RECEPTOR] [--rec_residue REC_RESIDUE]  
[--tether_smarts TETHER_SMARTS]  
[--tether_smarts_indices IDX IDX]  
mk_prepare_ligand.py: error: the following arguments are required: -i/--mol  
(vina) esh@debian:~$ vina  
AutoDock Vina v1.2.7  
#####  
# If you used AutoDock Vina in your work, please cite: #  
# #  
# J. Eberhardt, D. Santos-Martins, A. F. Tillack, and S. Forli #  
# AutoDock Vina 1.2.0: New Docking Methods, Expanded Force #  
# Field, and Python Bindings, J. Chem. Inf. Model. (2021) #  
# DOI 10.1021/acs.jcim.1c00203 #  
# #  
# O. Trott, A. J. Olson, #  
# AutoDock Vina: improving the speed and accuracy of docking #  
# with a new scoring function, efficient optimization and #  
# multithreading, J. Comp. Chem. (2010) #  
# DOI 10.1002/jcc.21334 #  
# #  
# Please see https://github.com/ccsb-scripps/AutoDock-Vina for #  
# more information. #  
#####
```

Тестовый запуск стыковки:

1. Скачать по ссылке <https://www.rcsb.org/structure/1IEP> файл в Legacy PDB формате



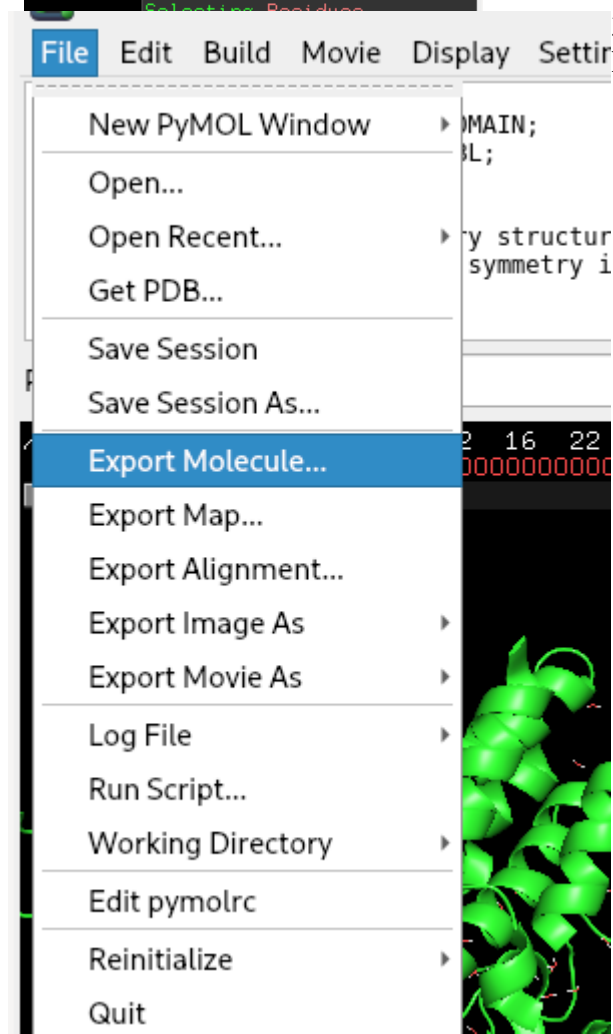
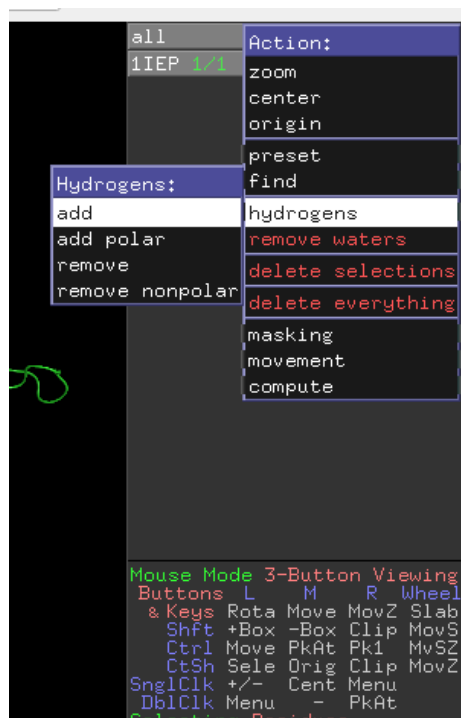
2. Перейти в загрузки cd «Загрузки» (кавычки должны быть верхние, но у меня они не печатаются в документе)

3. Далее через программу PyMOL либо через avogadro добавляем водороды

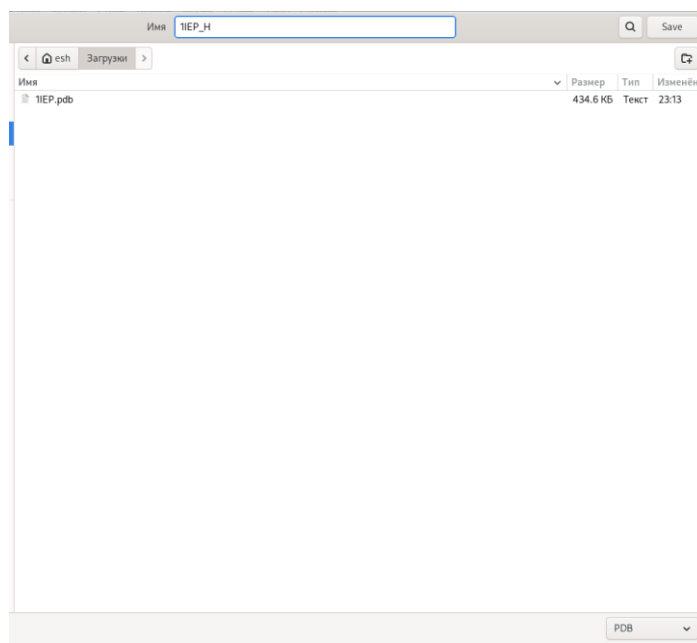
Добавление через PyMOL

1. открыть файл через pyMol

2. выбрать Action → hydrogens → add



3. file → export molecule → в формате выбрать PDB



4. Возвращаемся в терминал, вводим команду `mk_prepare_receptor.py -i 1IEP_H.pdb -o 1iep_receptor -p -v -d "НОН,SO4,PO4,CL,NA" --box_size 20 20 20 --box_center 15.190 53.903 16.917`, в результате будет создан файл формата `pdbqt`, в него не будут записаны лишние остатки.

```
(vina) esh@debian: ~/python$ mk_prepare_receptor.py -i 1IEP_H.pdb -o 1iep_receptor -p -v -d "НОН,SO4,PO4,CL,NA" --box_size 20 20 20 --box_center 15.190 53.903 16.917
@> 9484 atoms and 1 coordinate set(s) were parsed in 0.03s.
/home/esh/miniconda3/envs/vina/lib/python3.14/site-packages/meeko/polymer.py:856: RuntimeWarning: Input residues {'A:201': 'STI', 'B:202': 'STI'} not in residue_templates
  warnings.warn(err, RuntimeWarning)
/home/esh/miniconda3/envs/vina/lib/python3.14/site-packages/meeko/polymer.py:857: RuntimeWarning: Trying to resolve unknown residues by building chemical templates...
  warnings.warn(f"Trying to resolve unknown residues by building chemical templates... ", RuntimeWarning)
Molecule doesn't contain matching atoms for acidic proton loc: {'[H][O][CX3](=O)': 0, '[H][O][SX4](=O)': 0, '[H][O][SX3](=O)': 0, '[H][O][PX4](=O)': 0, '[H][O][PX3](=O)': 0, '[H][O][CX3](=S)': 0, '[H][O][SX4](=S)': 0, '[H][SX2][a]': 0}-> deprotonate returning original mol...
No template matched for residue key='A:1'
tried 1 templates for residue key='A:1'excess_H ok=False
CL      heavy_miss=0 heavy_excess=0 H_excess=[0] bond_miss=set() bond_excess=set()

No template matched for residue key='A:2'
tried 1 templates for residue key='A:2'excess_H ok=False
CL      heavy_miss=0 heavy_excess=0 H_excess=[0] bond_miss=set() bond_excess=set()

No template matched for residue key='A:4'
tried 1 templates for residue key='A:4'excess_H ok=False
CL      heavy_miss=0 heavy_excess=0 H_excess=[0] bond_miss=set() bond_excess=set()

No template matched for residue key='A:5'
tried 1 templates for residue key='A:5'excess_H ok=False
CL      heavy_miss=0 heavy_excess=0 H_excess=[0] bond_miss=set() bond_excess=[0]

matched with excess inter-residue bond(s): A:5
matched with excess inter-residue bond(s): A:253
No template matched for residue key='B:3'
tried 1 templates for residue key='B:3'excess_H ok=False
CL      heavy_miss=0 heavy_excess=0 H_excess=[0] bond_miss=set() bond_excess=set()

No template matched for residue key='B:6'
tried 1 templates for residue key='B:6'excess_H ok=False
CL      heavy_miss=0 heavy_excess=0 H_excess=[0] bond_miss=set() bond_excess=set()

Error: Creation of data structure for receptor failed.

Details:
  Template matching failed for: ['A:1', 'A:2', 'A:4', 'A:5', 'B:3', 'B:6']
  These residues can be ignored with option allow_bad_res.

Recommendations:
1. (for batch processing) Use -a/--allow_bad_res to automatically remove residues
   that do not match templates, and --default_altloc to set
   a default altloc variant. Use these at your own risk.

2. (processing individual structure) Inspecting and fixing the input structure is recommended.
   Use --wanted_altloc to set variants for specific residues.

(vina) esh@debian: ~/python$
```

5. если появилось такое предупреждение, то можно сразу удалить проблемные остатки с использованием флага `-a`:

`mk_prepare_receptor.py -i 1IEP_H.pdb -o 1iep_receptor -p -v -d "НОН,SO4,PO4,CL,NA" --box_size 20 20 20 --box_center 15.190 53.903 16.917 -a`

Здесь в самом низу указано, что файлы записаны. Ура!

```

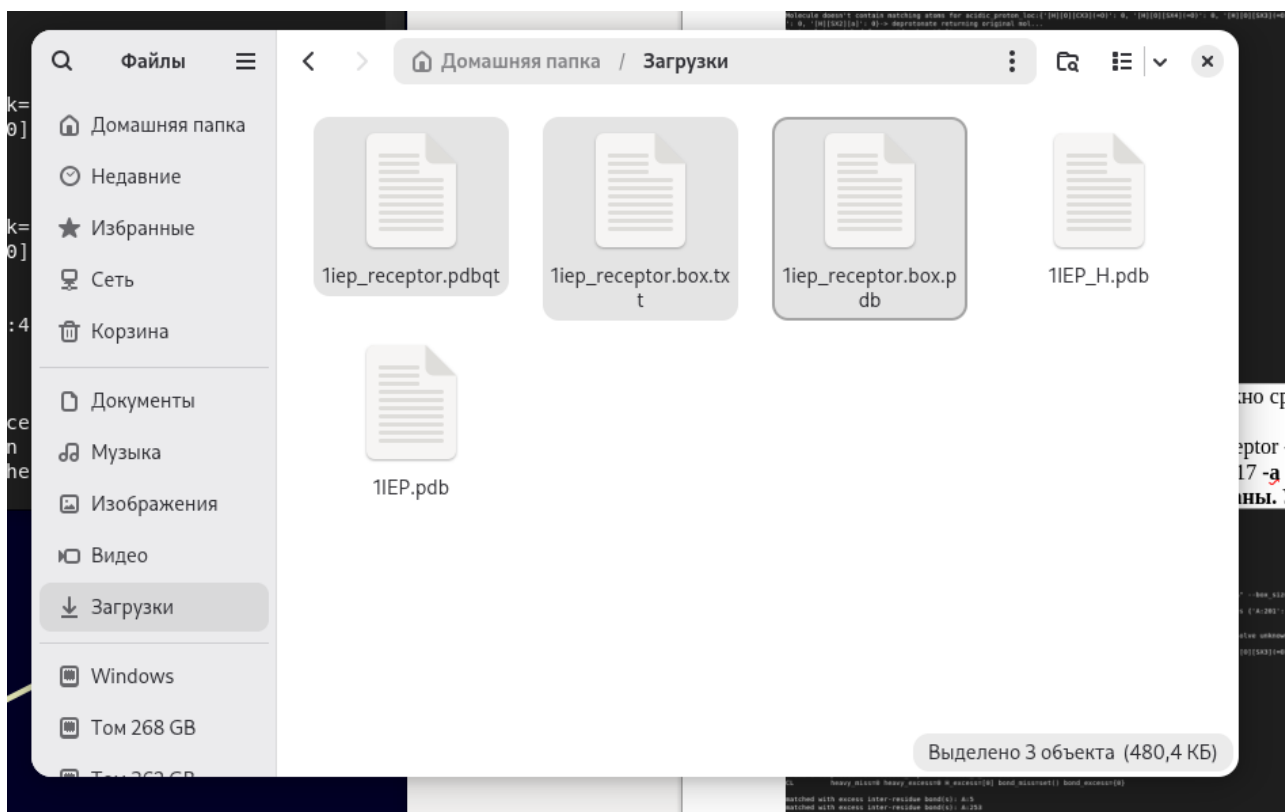
Recommendations:
1. (for batch processing) Use -a/--allow_bad_res to automatically remove residues
that do not match templates, and --default_atloc to set
a default atloc variant. Use these at your own risk.
2. (processing individual structure) Inspecting and fixing the input structure is recommended.
Use --wanted_atloc to set variants for specific residues.

(vina) esh@debian: ~/python3$ mk_prepare_receptor.py -i 1IEP_H.pdb -o 1iep_receptor -p -v -d "HOH,S04,P04,CL,NA" --box_size 20 20 20 --box_center 15.190 53.903 16.917 -a
No 9484 atoms and 1 coordinate set(s) were parsed in 0.03s.
/home/esh/miniconda3/envs/vina/lib/python3.14/site-packages/meeko/polymer.py:856: RuntimeWarning: Input residues {'A:201': 'STI', 'B:202': 'STI'} not in residue_templates
warnings.warn(err, RuntimeWarning)
/home/esh/miniconda3/envs/vina/lib/python3.14/site-packages/meeko/polymer.py:857: RuntimeWarning: Trying to resolve unknown residues by building chemical templates...
warnings.warn("Trying to resolve unknown residues by building chemical templates...", RuntimeWarning)
Molecule doesn't contain matching atoms for acidic proton loc: {'[H][O][CX3](=O)': 0, '[H][O][SX4](=O)': 0, '[H][O][SX3](=O)': 0, '[H][O][PX4](=O)': 0, '[H][O][PX3](=O)': 0, '[H][O][CX3](=S)': 0, '[H][O][SX4](=S)': 0, '[H][SX2][a]': 0}-> deprotonate returning original mol...
No template matched for residue_key='A:1'
tried 1 templates for residue_key='A:1'excess_H ok=False
CL heavy_miss=0 heavy_excess=0 H_excess=[0] bond_miss=set() bond_excess=set()
No template matched for residue_key='A:2'
tried 1 templates for residue_key='A:2'excess_H ok=False
CL heavy_miss=0 heavy_excess=0 H_excess=[0] bond_miss=set() bond_excess=set()
No template matched for residue_key='A:4'
tried 1 templates for residue_key='A:4'excess_H ok=False
CL heavy_miss=0 heavy_excess=0 H_excess=[0] bond_miss=set() bond_excess=set()
No template matched for residue_key='A:5'
tried 1 templates for residue_key='A:5'excess_H ok=False
CL heavy_miss=0 heavy_excess=0 H_excess=[0] bond_miss=set() bond_excess=0
matched with excess inter-residue bond(s): A:5
matched with excess inter-residue bond(s): A:253
No template matched for residue_key='B:3'
tried 1 templates for residue_key='B:3'excess_H ok=False
CL heavy_miss=0 heavy_excess=0 H_excess=[0] bond_miss=set() bond_excess=set()
No template matched for residue_key='B:6'
tried 1 templates for residue_key='B:6'excess_H ok=False
CL heavy_miss=0 heavy_excess=0 H_excess=[0] bond_miss=set() bond_excess=set()
- Template matching failed for: ['A:1', 'A:2', 'A:4', 'A:5', 'B:3', 'B:6'] Ignored due to allow_bad_res.

Files written:
1iep_receptor.pdbqt <-- static (i.e., rigid) receptor input file
1iep_receptor.box.txt <-- Vina-style box dimension file
1iep_receptor.box.pdb <-- PDB file to visualize the grid box
(vina) esh@debian: ~/python3$

```

В папке, в которой мы работаем, файлы также увидим



Теперь переходим к подготовке лиганда. Скачиваем файл лиганда и запускаем команду `mk_prepare_ligand.py -i 1iep_ligand.sdf -o 1iep_ligand.pdbqt`

```

(vina) esh@debian:~/Зарпузки$ mk_prepare_ligand.py -i liep_ligand.sdf -o liep_ligand.pdbqt
Input molecules processed: 1, skipped: 0
PDBQT files written: 1
PDBQT files not written due to error: 0
Input molecules with errors: 0
(vina) esh@debian:~/Зарпузки$

```

Далее переходим к запуску докинга (на сайте в инструкции это 4.b пункт, третий пункт мы пропускаем, поскольку он просто для AUTODOCK 4)

```

vina --receptor liep_receptor.pdbqt --ligand liep_ligand.pdbqt \
    --config liep_receptor.box.txt \
    --exhaustiveness=32 --out liep_ligand_vina_out.pdbqt

```

Если все успешно, то запустится докинг

```

(vina) esh@debian:~/Зарпузки$ vina --receptor liep_receptor.pdbqt --ligand liep_ligand.pdbqt \
    --config liep_receptor.box.txt \
    --exhaustiveness=32 --out liep_ligand_vina_out.pdbqt
AutoDock Vina v1.2.7
#####
# If you used AutoDock Vina in your work, please cite:      #
#                                                            #
# J. Eberhardt, D. Santos-Martins, A. F. Tillack, and S. Forli #
# AutoDock Vina 1.2.0: New Docking Methods, Expanded Force  #
# Field, and Python Bindings, J. Chem. Inf. Model. (2021)   #
# DOI 10.1021/acs.jcim.1c00203                               #
#                                                            #
# O. Trott, A. J. Olson,                                     #
# AutoDock Vina: improving the speed and accuracy of docking #
# with a new scoring function, efficient optimization and    #
# multithreading, J. Comp. Chem. (2010)                      #
# DOI 10.1002/jcc.21334                                       #
#                                                            #
# Please see https://github.com/ccsb-scripps/AutoDock-Vina for #
# more information.                                           #
#####
Scoring function : vina
Rigid receptor: liep_receptor.pdbqt
Ligand: liep_ligand.pdbqt
Grid center: X 15.19 Y 53.903 Z 16.917
Grid size : X 20 Y 20 Z 20
Grid space : 0.375
Exhaustiveness: 32
CPU: 0
Verbosity: 1

Computing Vina grid ... done.
Performing docking (random seed: -292489323) ...
0% 10 20 30 40 50 60 70 80 90 100%
|----|----|----|----|----|----|----|----|----|
*****

mode |  affinity | dist from best mode
    | (kcal/mol) | rmsd l.b. | rmsd u.b.
-----+-----+-----+-----
  1 |  0.02233 |      0    |      0
  2 |  0.7815  |    1.463  |    4.533
  3 |  1.267   |    1.489  |    4.339
  4 |  1.314   |    1.482  |    4.528
  5 |  3.696   |    1.434  |    2.019
  6 |  5.002   |    1.523  |    4.409
  7 |  8.398   |    1.537  |    2.222
  8 | 251.8    |    12.19  |    15.41
  9 | 322.9    |    2.285  |    9.977
(vina) esh@debian:~/Зарпузки$

```

Если оценивать результаты докинга, то здесь всё плохо, так как в структуре были лишние элементы и лишняя молекула. Но как проверка работоспособности — вполне себе.

Рабочие файлы для проверки можно взять на гитхабе - https://github.com/ccsb-scripps/AutoDock-Vina/tree/develop/example/basic_docking/data